

MILESTONE 0

HOW WE GO ONLINE

NGO for the live table. Firebase for accounts and matchmaking.

LIVE TABLE

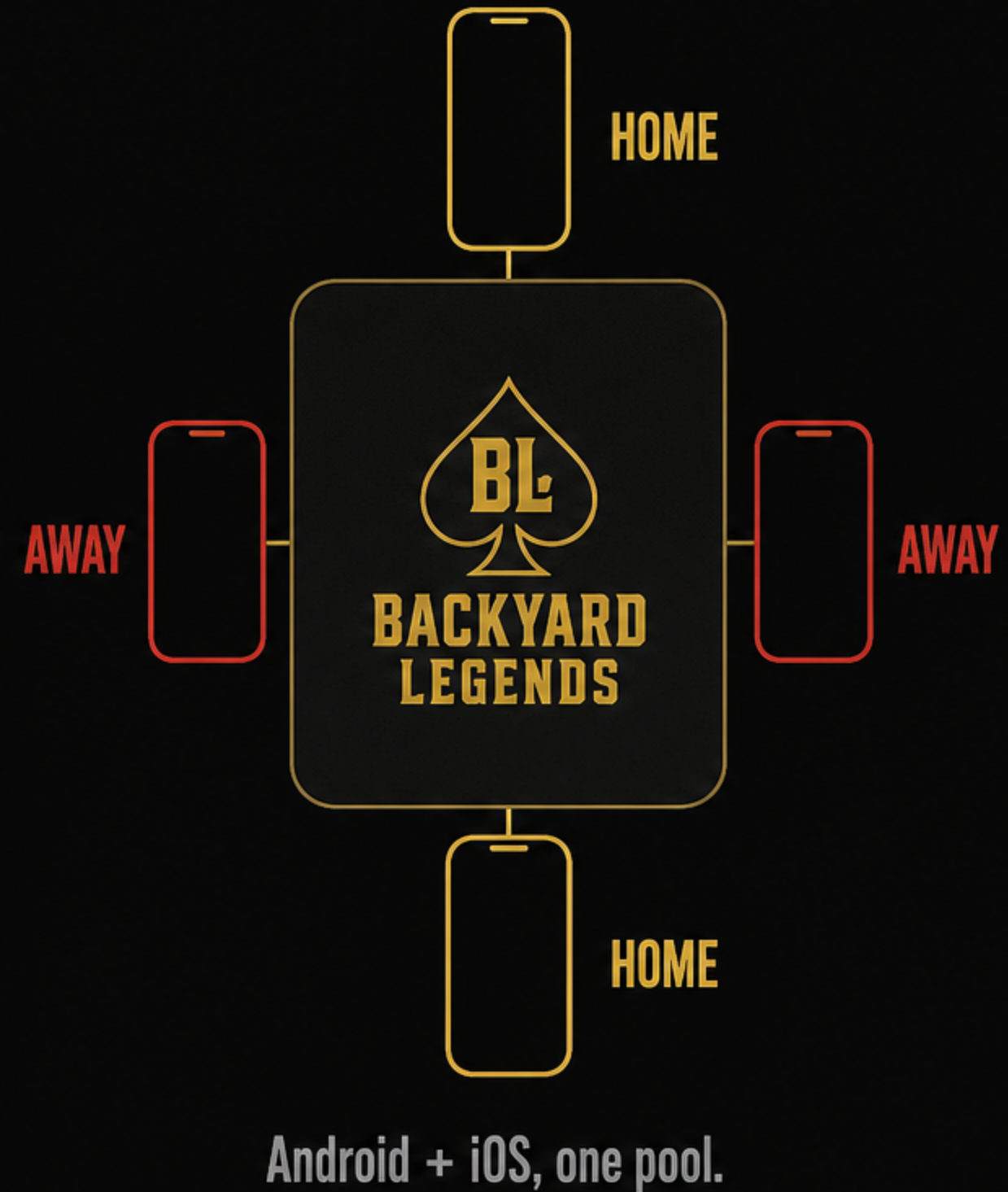
Unity NGO 2.13 — real-time 4-player 2v2

CONNECT

Unity Relay — mobile NAT only

BACKEND

Firebase — Auth, Firestore, Functions, ranking



Host-authoritative SpadesMatchController. No dedicated game servers in this phase.

What each system owns

UNITY NGO (LIVE)



Host runs SpadesMatchController



Shuffle, deal, bids, tricks, score



Private hands stay on host



Clients send Bid / Play / Claim / Forfeit



Opponents see card backs only

snapshot each action



FIREBASE (DURABLE)



Auth: Apple, Google, anonymous



Profiles, rating, invite codes



Table session + public snapshot



Matchmaking Cloud Functions



Match result and history



Unity Relay carries NGO packets.

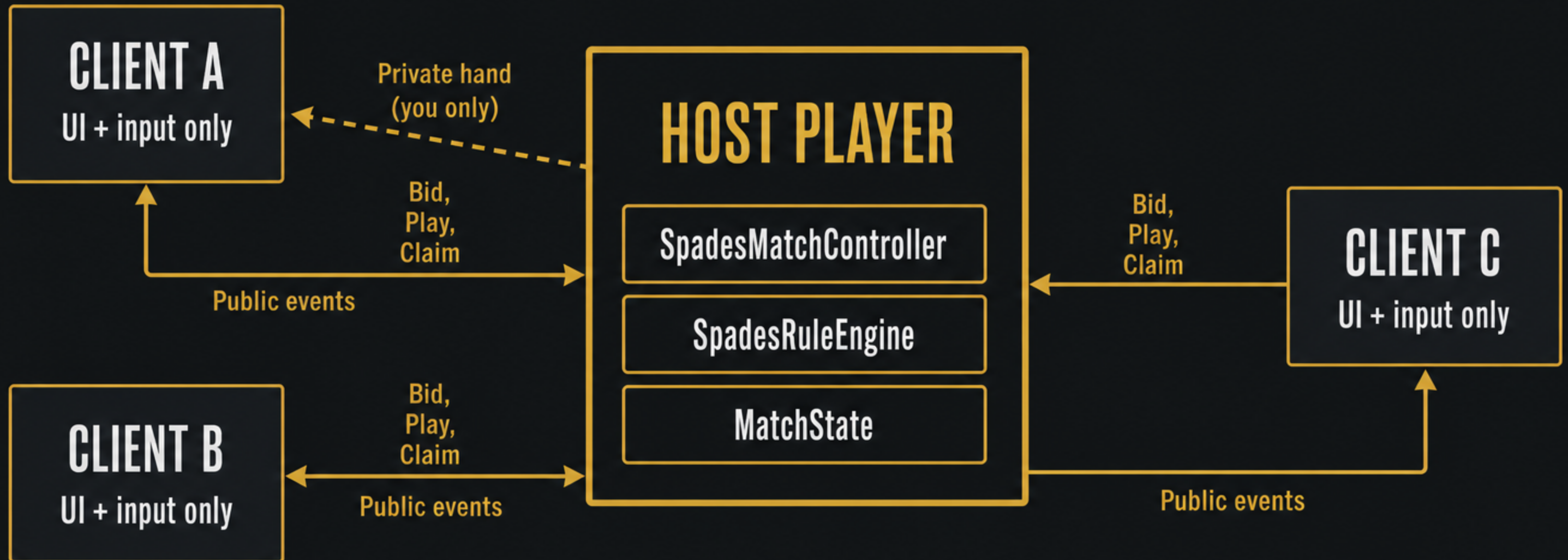
Firestore does not replace Relay.

FROM LOGIN TO THE TABLE.

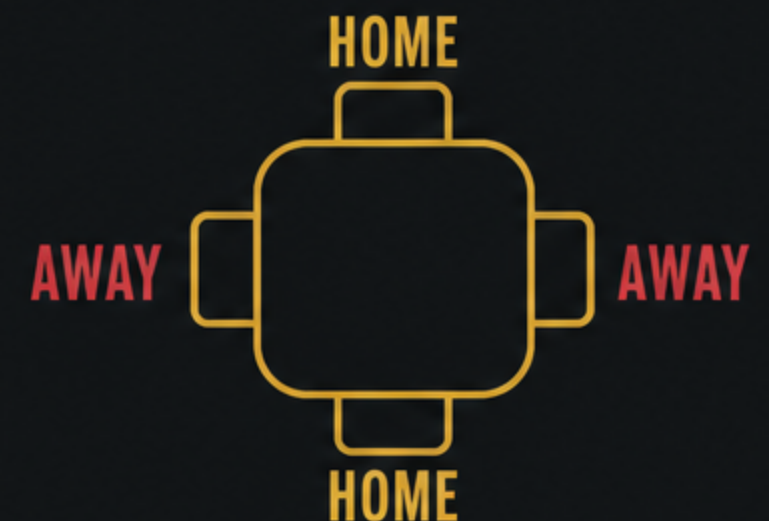


Private table uses invite code. **Ranked** uses the same table after matchmaking.

Host is the referee.



Illegal moves rejected by host. Same rules as Phase 1 local game.



DISCONNECT AND RECONNECT



PLAYER DROPS (HOST STILL LIVE)

- 1 | TABLE PAUSES
- 2 | 90 SECOND GRACE, VISIBLE TO ALL
- 3 | PLAYER RETURNS: CATCH-UP, NO PENALTY
- 4 | GRACE ENDS: AI SITS IN, MATCH CONTINUES
- 5 | ORIGINAL PLAYER MAY RECLAIM UNTIL MATCH END



HOST DROPS

- 1 | TABLE PAUSES, 90 SECOND GRACE
- 2 | HOST RETURNS: SAME TABLE RESUMES
- 3 | GRACE ENDS: PROMOTE A NEW HOST
- 4 | RESTORE FROM FIRESTORE SNAPSHOT
- 5 | PLAYERS JOIN A NEW RELAY



RANKED FORFEIT-ON-TIMEOUT IS LATER, NOT THIS PHASE. UNRANKED USES AI SIT-IN.
GRACE IS 90 SECONDS.

Three rules locked.

Client-approved for Milestone 0

CLAIM THE REST

- Live play only
- Claiming team cards revealed
- Both opponents must accept
- Either reject: hand continues
- Both accept: books awarded, score now

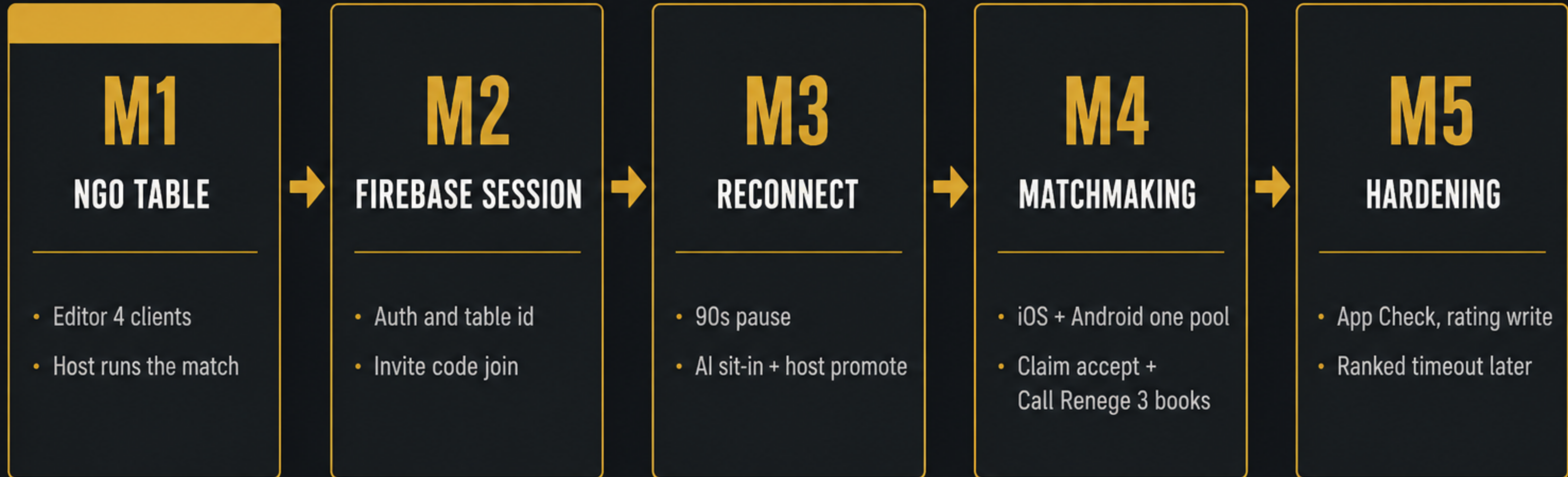
CALL RENEGE

- Street Mode only
- Opponent calls renege
- Upheld: guilty team loses **3 books**
- Those **3 books** go to opponents
- Not a minus 200 score penalty

DISCONNECT

- Table pauses **90 seconds**
- Return: resume, no penalty
- Unranked miss: AI sits in
- Host miss: promote new host
- Forfeit button: opponents win now

BUILD ORDER AFTER APPROVAL.



PHASE 1

PHASE 1 CORE

Editor 4 clients

Firebase session + invite

90s reconnect

iOS + Android one pool

Claim accept +
Street Call Renege 3 books

Hardened baseline

CLAIM: ACCEPT CLAIM OR PASS

STREET CALL RENEGE: 3 BOOKS